# David Liu

*ML Research Engineer & Senior Data Scientist · 14 years writing code, a decade of production ML across biotech, e-commerce, healthcare, recruiting, and fintech.*

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## SUMMARY

At every company I've worked for, I have built production machine learning systems end-to-end — data ingestion, feature engineering, training, serving, and measurement — and I instrument them to verify their own claims. My work has a consistent pattern: question the accepted metric, find the architectural gap, build the system others extend into production. Currently leading personalization at Shipt for millions of grocery shoppers, and concurrently the lead AI engineer at a healthcare stealth startup, where I architected the clinical-scoring AI framework end to end, from document intake and scoring to clinician review and evaluation. Previously a Machine Learning Research Engineer at Freenome (multiomics cancer detection) and a Sr. ML Engineer at Change Healthcare. I also run an independent AI research program — three papers spanning multi-agent synthesis, model selection under distribution shift, and LLM-grounded feature extraction — with evaluation designed so the system that produces a number is never the one that grades it.

## EXPERIENCE

### Lead AI Engineer — Stealth Startup (Behavioral Health AI)

MAR 2026 — PRESENT

*Lead AI engineer (concurrent with Shipt) at a healthcare startup building behavioral-health solutions with LLMs. I architected its clinical-scoring AI framework end to end, from document intake to clinician review.*

- Architected the production scoring pipeline: OCR on case documents, an LLM on **AWS Bedrock** (zero-retention) scoring each clinical criterion, and a rule-based tree that rolls criteria up to the final score. Quoted evidence is checked against the source text, so every score traces back to the chart.
- Designed the trust layer: recommendation lanes (auto-release, suggest, review-required, suppress), fixed safety gates for high-risk findings, and reviewer-reliability weighting. Output is advisory; the **clinician always has the final say**.
- Built the evaluation loop: a run ledger, A/B arms with significance tests, and scorecards against clinician-reviewed cases. A change ships only if it passes a no-net-regression gate; most prompt changes I tested were rejected on that evidence. Same expert-gated pattern as my **Adaptive Domain Intelligence** research.
- Designed a model adapter and response cache for like-for-like model comparison; outputs are validated in a **PHI-compliant AI Sandbox**.
- Built an **LLM-as-judge** pipeline for visual elements in clinical documents, and SMOTE-inspired synthetic test cases (controlled perturbations of real examples) to widen regression coverage.

### Senior Data Scientist — Shipt (Personalization Team)

SEP 2024 — PRESENT

*Lead data scientist on Shipt's Personalization team; designed and built the recommendation platform behind personalized shelves for millions of shoppers, from model to measurement, with Engineering on serving. Mentored 4 data scientists, including 2 direct reports.*

- Built and launched Deals For You V2, replacing matrix factorization with two-tower retrieval and precision reranking; a CUPED-corrected A/B test drove **+91% shelf engagement**, **+1.76% order incidence**, and **+1.41% orders per user**.
- Audited pre-treatment imbalance and a variant activation gap before SVP planning, correcting the annualized estimate from +83K to **~50K incremental orders (95% CI: 20K–80K)** and **~$3.7M directional GMV**; preserved a real win while retiring an inflated topline.
- The validated result prompted leadership to re-plan H2 around Deals as a potential **~$6M contributor to Shipt's $20M incremental-GMV target**, prioritizing scale-up, holdback measurement, and reuse of the eligibility platform.
- Architected a reusable real-time Eligibility platform: a store-scoped hard pre-filter before ANN with swappable content tables; reduced an initial **448M-row design by ~95%** to ~20M store-specific rows plus a compact global set, zero-downtime refreshes, and a p99 <750 ms serving target.
- Across the broader personalization portfolio, delivered up to **23% Personalized GMV lift** across A/B-tested shelves and a cold-start strategy with **+47% engagement** and **+8% first-time orders**.
- Created the Personalization Interaction Score (views → clicks → ATC → purchases) and an automated Shelf Attribution pipeline; challenged an irreproducible 5% attribution claim, established the defensible 1–4% range, and raised the organization's measurement standard.
- Partnered with Engineering to replace CSV-based recommendation delivery with Kafka pipelines across legacy recommenders; served as the team's sole data scientist for its first four months while maintaining 16 repositories.
- Designed a Retrieval-Augmented Generation system over Shipt's retailer catalogs and built the business case that secured investment in an agentic-AI framework.

### Machine Learning Research Engineer — Freenome

NOV 2020 — JUN 2024

*Core ML engineer at a genomics company developing a blood test for early-stage cancer detection. I worked across the whole model lifecycle, from data and training to serving and audit.*

- Key contributor to Freenome's core multiomics cancer-detection model that predicts cancer stage (1–4) from blood-draw data; built data abstractions for petabyte-scale genomic datasets that unblocked cross-analyte feature development and accelerated training/evaluation cycles.
- Built a model-comparison system tracking research-vs-production model performance side-by-side — required for FDA audit compliance, gave the team confidence that production models stayed aligned with research intent.
- Designed and built large portions of Freenome's distributed ML training and serving platform used daily by 30+ scientists; scaled CPU-bound training across O(100) machines, supported leave-one-out / K-fold evaluation, and built reproducible model-artifact storage.
- Led adoption of PyTorch, MLFlow, and RayTune across the ML team — replaced legacy tooling to unlock GPU acceleration, experiment tracking, and hyperparameter tuning at scale.
- Built a cloud-cost monitoring system surfacing the biggest GCP storage and compute expenses; visibility alone drove optimizations saving the company **over $10M annually** — one of the highest-ROI projects I've worked on.
- Recognized with a Servant Leadership Award, elected by managers and peers across the engineering organization.

### Sr. Machine Learning Engineer — Change Healthcare

JAN 2020 — NOV 2020

*ML systems for health-insurance claims processing, where accuracy translates directly into operational cost savings. I built the models and their path to production.*

- Designed a ranking model matching human workers to claims tasks based on skill, history, and complexity; **$7M annual value** by reducing manual task assignment and the need for additional hires.
- Built a classification model partitioning sensitive patient documents (image + text) to route claims to the correct processing workflow.
- Developed internal AWS tooling and production API infrastructure for the ML team's deployment pipeline.
- Led a cross-functional tiger team prototyping a conversational chatbot (Rasa + HuggingFace NLP) for internal claims-inquiry workflows.

### Data Scientist — Riviera Partners

JAN 2019 — DEC 2019

*ML models for an executive-recruiting firm; full pipeline from data collection to model serving.*

- Developed a model suite: a job-departure-likelihood classifier, a regression model predicting team sizes from resume features, and a candidate-ranking model using a custom NDCG listwise loss.
- Built an end-to-end framework for rapid model prototyping, training, evaluation, and serving — enabled the team to iterate on new models without re-engineering infrastructure each time.
- Wrote scrapers harvesting structured candidate data from public sites and APIs.

### Undergraduate Researcher — UC Berkeley

JAN 2017 — DEC 2018

- **California Institute for Energy and Environment (CIEE)** — built a recurrent neural network for predicting building energy usage, exploring how temporal consumption patterns can inform smarter grid management.
- **Bengson Research Lab, Sonoma State** — applied ML to EEG data to predict individualized occipital lobe activation patterns; demonstrated early feasibility for brain-computer interface applications.

### Earlier Roles

2015 — 2017

- **Data Science Intern, Castlight Health** (2017) — entity matching and deduplication pipeline using gradient-boosted classifiers with hard-negative mining; 85–95% precision/recall across hospital, facility, and practitioner entities.
- **Data Science Contractor, Riviera Partners** (2016) — team-size prediction model from public data; Python wrapper for survival-model time-series analysis; Flask model-serving infrastructure.
- **URAP, Berkeley Institute of Data Science** (2016) — mapped UC Berkeley course progression across majors via class-taxonomy organization and deduplication.
- **Data Science Intern, Doximity** (2015) — gradient-boosted classifier for malformed scraped articles; reverse geocoding + fuzzy string matching to link doctors in news articles to facility profiles.

## SELECTED PROJECTS

### Adaptive Domain Intelligence — LLM-as-Feature-Engineer Architecture

2026 · RESEARCH · WORKING PAPER

Research architecture for compressing qualitative records (court opinions, clinical notes) into calibrated predictions without trusting the LLM: the model only proposes feature schemas and performs verbatim-quote-grounded extraction — unsupported values are dropped and counted toward a measured hallucination rate — while a small calibrated classical model makes the prediction; self-improvement is gated by a domain expert and a frozen gold set (permutation test + subgroup-fairness veto). Validated as a known-answer test on four real legal corpora: re-derives six decades of established judicial-politics signals and nulls, and correctly lands at chance on an ECtHR leakage trap where prior text-only work reported ~79%. An automated verifier re-derives every headline number (8/8, zero drift).

### Meta Council — Multi-Expert AI Decision Support Platform

2025 — PRESENT · RESEARCH & PRODUCT

Multi-agent LLM framework where N expert agents (each with a unique persona and analytical framework) analyze queries in parallel, then a weighted synthesis step produces structured decision documents with confidence scores, dissent preservation, and risk matrices. Evaluated across 750+ benchmark runs spanning 6 domains and 5 models (3B to frontier-class): synthesis outperforms single-best by 29–58% ($p<0.0001$, $d=2.16$); the optimal aggregation method is domain-dependent; synthesis amplifies model quality non-linearly. Published as an independent research paper.

### AutoTrader — ML-Powered Stock Prediction System 2024 — PRESENT · PERSONAL

Fully autonomous market-prediction system: collects nightly market data for 600+ tickers, engineers 500+ features across 8 source families, trains 1,800+ dual models (classifier for direction, regressor for magnitude), and delivers confidence-ranked predictions before market open every weekday. Built every piece — data ingestion, custom feature store, dual-model training framework with walk-forward validation and Optuna, FAISS-powered similarity search, tiered email subscription system with Stripe billing, multi-cloud GCP+Azure infrastructure — running autonomously for ~$235/month.

### Patterns of Choice — Revealed vs. Stated Ethical Values 2026 · RESEARCH · OPEN-SOURCE

Longitudinal instrument measuring the gap between the ethical values people claim and the values their choices reveal, built around a pre-registered hypothesis that recurring-character narrative immersion reduces social-desirability bias. Shipped as a local-first offline PWA — event-sourced engine, deterministic on-device scoring, nothing leaves the device — with a 61-scenario corpus, five recurring-character arcs, and a cross-implementation parity gate between the app runtime and the analysis pipeline.

### XGBoost Visual Guide 2026 · OPEN-SOURCE

Interactive visual textbook explaining XGBoost and gradient boosting from first principles — 10 sections covering decision trees, ensemble methods, animated step-by-step gradient boosting, learning rate effects, XGBoost-specific innovations (histogram splits, sparsity handling), early stopping, feature importance, and a hyperparameter cheat sheet. D3.js + Chart.js.

## PUBLICATIONS

**Adaptive Domain Intelligence: An LLM-as-Feature-Engineer Architecture for Compressing Qualitative Records into Calibrated Empirical Predictions**

*2026 · Independent Research · Working Paper · LLM feature engineering, grounded extraction, hallucination measurement, self-improving systems, calibration, legal NLP*

**Weighted Multi-Expert Synthesis for High-Stakes Decision Support: A Multi-Agent LLM Framework with Dissent Preservation**

*2026 · Independent Research · CC BY 4.0 · Multi-agent systems, LLM, decision support, weighted synthesis, dissent preservation, confidence calibration*

**Stability Bonus Regularization for Model Selection Under Positive-Class Distribution Shift**

*2026 · Independent Research · CC BY 4.0 · Model selection, distribution shift, cross-validation, class imbalance, regularization*

## EDUCATION

**University of California, Berkeley** — BS, Computer Science and Data Science (Dual Degree), Class of 2018. Berkeley Institute of Data Science Undergraduate Research Apprenticeship (2015).

## TECHNICAL SKILLS

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| **LANGUAGES** | Python · SQL · Bash · Java · C/C++ |
| **ML & DATA** | PyTorch · XGBoost · Scikit-learn · Pandas · MLFlow · FAISS · Gensim · NLTK · SpaCy |
| **LLM & AGENTS** | Claude & GPT APIs · AWS Bedrock · multi-agent orchestration · RAG · grounded extraction · LLM-as-judge · LLM evaluation & hallucination measurement · Langfuse · agentic coding workflows |
| **EVALUATION & TESTING** | evaluation harnesses & known-answer tests · benchmark design · A/B testing · prompt experimentation · synthetic test-case generation · walk-forward validation · permutation tests · calibration · pre-registered hypotheses |
| **CLOUD** | GCP · AWS · Azure |
| **DATA INFRASTRUCTURE** | PostgreSQL · Snowflake · MySQL · Spark · Kafka |
| **ORCHESTRATION** | Airflow · Flyte · Metaflow · GitHub Actions |
| **INFRASTRUCTURE** | Docker · Kubernetes · Git · CI/CD |